

Tic Tac Toe online multiplayer game

#	Description	Functional /non functional	Testing input	Expected output	Pass
1	Ability to connect to the server.	Functional	User enters correct ip address and port number then clicks connect	We can see that both players have successfully joined the server once the game says who is player X and player O after assigning it and displaying it in the JTextArea made for indicating which symbol the user is assigned.	
2	Updates are made to GUI's tic tac toe board after player makes a move	Functional	A user makes a move by clicking one of the cells on the 3x3 board	Both GUIs for the clients are updated appropriately after move is made - the spot chosen by user is filled with X or O and GUI displays who's turn it is next in JTextArea dedicated to that	
3	Board state and board buttons on both GUIs are able to be "reset" so that they're empty and clickable	Functional	User clicks the reset game JButton OR the new game JButton	Clear all the input that was loaded into the code that stores info about tictactoeboard based on moves made by users during gameplay. Reset all elements in the code to be null/ clickable instead of filled with either X or O	
4	Player is able to make a move, and this	Functional	User makes a move by clicking an open space on the 3x3	If a player makes a move, it updates the cell clicked by	

	move is stored in code which allows for checking win or draw conditions		board	whichever symbol in tictactoeboard, effectively storing this update in the game board. Storing this info will help us know what constitutes a win or draw once further moves are made in game.	
5	Checks for game winner	Functional	User plays the game and gets 3 X or O symbols in a row. Then code should evaluate the win and say which player it is that won	The JTextArea on GUI is updated to either "player X wins" or "player O wins" after 3 of either symbol is in a row	PASS
6	Checks if the game is a draw	Functional	Users play the game in such a way that all spaces are filled but no 3-in-a-row combinations for X or O are found in the board	The JTextArea on GUI is updated to "No Winners, Game is a Draw"	PASS
7	Display End of Game after a win made	Functional	Users play game until completion (and a win or draw is declared after)	Make sure at the end it display certain person has won and prompts user to start a new game through use of "new game" JButton	PASS
8	Display the last move made on board by either player (ex: player X move to row 1 col 3)	Functional	Users make a move to any cell on the game board	The JTextArea that displays the last move played is updated to show which cell in the 3x3 board was selected and by which player after every move.	PASS
9	Reading thread keeps taking incoming	Non functional	Keep a count to see how long it takes for each player to receive the other player's	ensure that in 10 sec that the players are able to see each move the opposite player is	PASS

	messages/moves from other players		move after the other player has completed their turn	making via server updates to their local GUIs	
10	The server is up and running, able to be connected to by clients.	Non functional	Users are able to connect and disconnect from the localhost server in 10 sec.	The player is continuously connected to the server as long as the server is running. If you disconnect from the server you disconnect from the game and have no updates to your local GUI on your JFrame anymore.	PASS